

TrES-2b

R-1456

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130629 · created 2026-08-01T10:11:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456472.71971 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.043
Transit depth (Rp/Rs)^2	2.17 +/- 1.27 [%]
Orbital Inclination (inc)	84.5 +/- 2.91
Airmass coefficient 1 (a1)	0.52 +/- 0.18
Airmass coefficient 2 (a2)	0.54 +/- 0.46
Scatter in the residuals of the lightcurve fit is	35.9 %
Best Comparison Star	#2 - [572, 200]
Optimal Aperture	8.22
Optimal Annulus	19.87
Transit Duration (day)	0.071 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.043 vs 0.1254 (deviation 0.47σ)
Duration fit vs published	0.071 d vs 0.0746 d (deviation 0.08σ)
Mid-time O–C	-0.3 min
Depth SNR	3.2
Residual scatter	35.9 %

Light curve

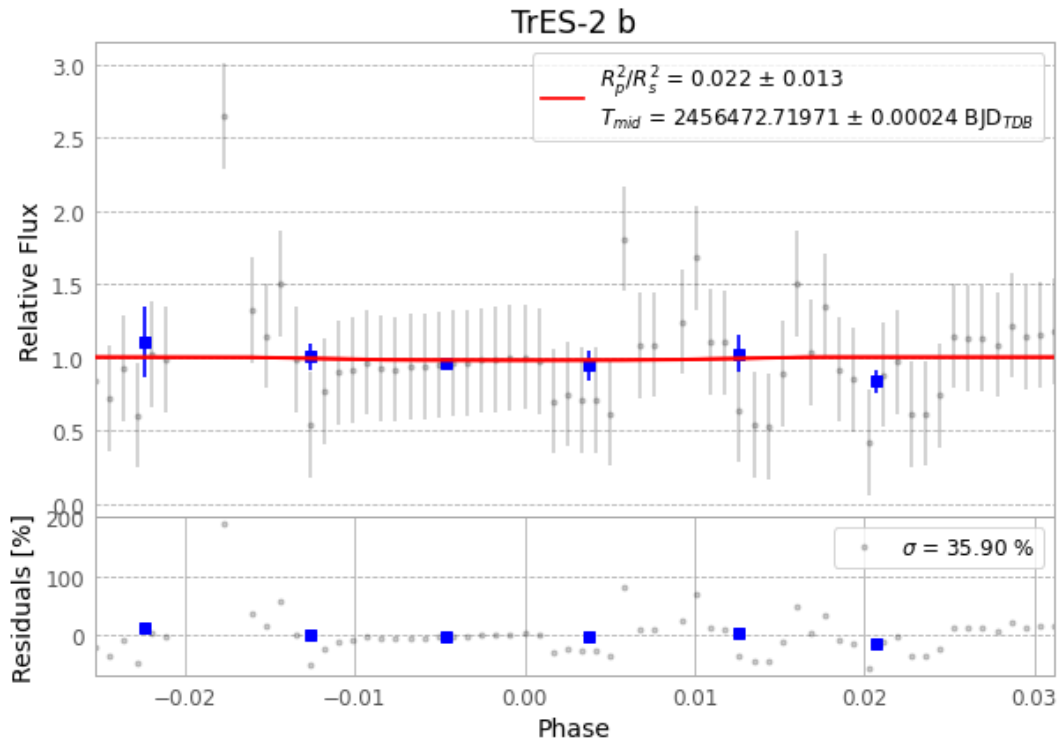


Figure 1 — FinalLightCurve_TrES-2 b_2013-06-28.png (SHA-256 e2213695ed892906...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 414], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 adc0be3ad9c269f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), TRES-2130629034212.FITS → TRES-2130629070313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130629.log (5cf2ef3516cd...), FinalLightCurve_TrES-2 b_2013-06-28.png (e2213695ed89...), FinalLightCurve_TrES-2 b_2013-06-28.csv (380ee17b2f3f...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 10:11Z.